INCH-POUND

MIL-L-10867E

29 March 1991
SUPERSEDING
MIL-L-10867D
17 December 1970

MILITARY SPECIFICATION

LEATHER, CATTLEHIDE, GUSSET, CHROME TANNED, FATLIQUORED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers chrome tanned, fatliquored cattlehide leather for gussets.

1.2 Classification. The gusset leather shall be of one type in the following treatments as specified (see 6.2).

Treatment A - Treated for resistance to water absorption
Treatment B - Treated for resistance to water penetration
Treatment C - Not treated for resistance to water

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8330

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-43432 - Packaging of Leather

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 22709.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Standard sample. The leather, when embossed, shall match the standard sample for fine hair cell pattern and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Material. The leather shall be chrome tanned cattlehide sides. The flesh side shall be smooth and free of loose flesh.

3.4 Trim (form) and area. The leather sides shall be trimmed to conform with the definition of a "side" as specified in FED-STD-311. The area expressed in square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.5 Thickness. The thickness of the leather shall be a minimum of 2-3/4 ounces and a maximum of 4.0 ounces when examined as specified in 4.4.3.

NOTE: 1 ounce = 1/64 inch.

3.6 Color. Unless otherwise specified (see 6.2), the color of the leather shall be black. The leather shall be either drum dyed or dyed on the grain side only.

3.6.1 Staining. When the leather is drum dyed, the stain on the test pad shall have an expanded AATCC rating of not less than 3 when the leather is tested as specified in 4.4.4.

3.7 Finish. The leather shall be buffed or snuffed only to the extent that it is necessary to remove superficial blemishes. The leather shall be finished on the grain side with a light application of finish and dye containing only sufficient pigment to assist in making the color uniform. The dye and finish shall be of the same color. The finish of the grain side shall be smooth. A fine hair cell embossed pattern may be applied to the grain surface of the side. Printed leather shall not be permitted. If embossed, the leather shall match the standard sample for fine hair cell pattern. Leather previously dyed or finished in a color not specified and then redyed or refinished to match a specified color shall not be acceptable.

3.8 Paranitrophenol. Unless otherwise specified (see 6.2), the leather shall contain not less than 0.18 and not more than 0.70 paranitrophenol fungicide when tested as specified in 4.4.4.

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3.9 Water resistant treatment. When treatment A or B is specified (see 6.2), the leather shall be treated with a water resistant compound (see 6.6), and shall meet the following requirements as applicable.

3.9.1 Treatment A (water absorption). When treatment A is specified (see 6.2), specimens of the treated leather shall gain not more than 30 percent of their weight by water absorption when tested as specified in 4.4.4.

3.9.2 Treatment B (water penetration). When treatment B is specified (see 6.2), at least 80 percent of the specimens tested shall withstand not less than 100 taps of 2 pounds thrust without water penetration when tested as specified in 4.4.4.

3.10 Stitch tearing strength. At least 80 percent of the specimens tested shall have stitch tearing strength values of not less than 30 pounds when tested as specified in 4.4.4.

3.11 Shrinkage. The leather shall not shrink before the temperature reaches 98°C, nor shall the leather shrink when subjected to a temperature of 98°C ± 0.5°C for a period of 30 seconds when tested as specified in 4.4.4.

3.12 Stiffness. At least 80 percent of the specimens tested shall have a load scale reading of not more than 20 when tested as specified in 4.4.4.

3.13 Chemical requirements. The leather shall conform to the chemical requirements in table I when tested as specified in 4.4.4.

TABLE I. Chemical Requirements

Characteristics	Minimum	Maximum
Chloroform soluble material, percent: 1/		
Treatment A or B	10.0	25.0
Treatment C	10.0	20.0
Chromic oxide, hide substance basis, percent 1/	4.0	7.5
pH value	3.3	---

1/ Calculated on a moisture free basis.

3.14 Workmanship. The finished leather shall conform to the quality of product established by this specification, and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested as specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with the referenced documents.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.2 End item visual examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of sides. The sample unit shall be one side. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for this examination shall be performed in accordance with FED-STD-311. Separate examinations shall be performed for visual and dimensional characterisitics, but the same sample may be used for each examination.

Examine	Defect
Quality of leather	Not soft and pliable Coarse loose fiber on flesh side Stain or discoloration 1/ Boney, boardy or flanky leather Any cut, hole, brand, hip mark, scratch, heavy vein, deep fat wrinkle or surface imperfections which individually or collectively cause more than a 12 percent loss in the total area of the side 1/
Trim	Not as specified (i.e., not full sides)
Color	Not as specified
Finish	Not finished grain out Grain surface not smooth Color not uniform Embossing other than a fine hair cell pattern Printed design on leather Does not match standard sample for fine hair cell pattern
Marking	Omitted, illegible, or not in area specified

1/ Light well healed scratches or grub damage, light fat wrinkles, or slight stains shall not be classified as defects.

4.4.3 End item dimensional examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of sides. The sample unit shall be one side. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for this examination shall be performed in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more sample units. Separate examinations shall be performed for visual and dimensional characterisitics, but the same sample may be used for each examination.

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Examine	Defect
Area expressed in terms of square feet	Less than area marked on leather
Thickness 1/	Less than 2-3/4 ounces or greater than 4.0 ounces

1/ Thickness shall be determined in accordance with Method 1011 of FED-STD-311, except that no individual measurement shall be less than the minimum or greater than the maximum specified.

4.4.4 End item testing. The methods of testing specified in FED-STD-311, wherever applicable and as listed in table II, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

a. More than three specimens fail requirements for water penetration.
b. More than three specimens fail requirements for stitch tearing strength.
c. More than three specimens fail requirements for stiffness.
d. One or more specimens fail any of the remaining requirements applicable to the sample unit.
e. Any composite test result fails to meet the specified requirement.

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite
Material identification	3.3	1/	-	-
Tannage	3.3	1/	-	-
Staining (drum dyed leather)	3.6.1	3021 2/	X	-
Paranitrophenol	3.8	6711	-	X
Water absorption (treatment A)	3.9.1	8112	X	-
Water penetration (treatment B)	3.9.2	8121	X	-

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TABLE II. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite
Stitch tearing strength	3.10	2151	X	-
Shrinkage	3.11	7011	X	-
Stiffness	3.12	4.5.1	X	-
Chloroform soluble material	3.13	6341	-	X
Chromic oxide	3.13	6515	-	X
pH value	3.13	6621	-	X

1/ A certificate of compliance shall be submitted and will be acceptable for the requirement stated.

2/ Except that the flesh side of the leather shall be in contact with the wet pad.

4.4.5 Packaging inspection. The sampling and inspection of the preservation, packaging, palletization, and marking shall be in accordance with the requirements of MIL-P-43432.

4.5 Methods of inspection.

4.5.1 Stiffness. The specimen shall be 2 by 1 inches cut with the long dimension perpendicular to the backbone. The stiffness of the specimen shall be determined as specified in Method 4211 of FED-STD-311, using a 1/2 inch span and a total moment weight equivalent to 0.25 inch-pound. The specimen shall be clamped firmly in the vise and the apparatus engaged until the specimen has been subjected to an angular deflection of 40 degrees. The apparatus shall be immediately readjusted by hand so as to completely remove the load on the specimen. Then the specimen shall be subjected to an angular deflection of 20 degrees following the procedure specified in Method 4211. The stiffness shall be recorded as the load scale reading.

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5. PACKAGING

5.1 Preservation, packing, palletization and marking. The preservation, packing, palletization and marking for each level of protection shall be in accordance with the applicable requirements of MIL-P-43432 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The leather is intended for use in Department of Defense field, service, and general purpose footwear.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Treatment required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Color when black is not required (see 3.6).
f. When the fungicide paranitrophenol is not required (see 3.8).
g. Levels of preservation and packing (see 5.1).
h. When palletization is required (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Selection of hides. It is recommended that thin spready hides be used. Hides which can be practicably split to 3 ounces are desired.

6.6 Water resistant treatment. Silicone compounds have been found to be capable of imparting the required water resistant properties to the leather.

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6.7 Subject term (key word) listing.

Boots
Chrome tanned
Fatliquored
Hides
Shoes

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing Activity:

Army - GL

(Project 8330-0191)

Review activities:

Army - MD
Air Force - 11, 82
DLA - CT

User activity:

Navy - MC

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-10867E	1991 March 29

3. DOCUMENT TITLE
Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP, APR 90 *Previous editions are obsolete.* 195/290

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